

# Lorentzian propinquity convergence on truncated $SU(2) \times U(1)_T$ Krein spectral triples

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## Abstract

We construct a Lorentzian analog of the Latrémolière quantum Gromov–Hausdorff propinquity, restricted to the Krein-positive state space, and prove the corresponding convergence theorem on truncated Krein spectral triples over the compact-temporal Wick-rotated manifold  $S^3 \times S_T^1$ . Concretely, let  $\mathcal{K}_{n_{\max}, N_t} := \mathcal{H}_{\text{GV}}^{n_{\max}} \otimes \mathbb{C}^{N_t}$  be the chirality-doubled Camporesi–Higuchi spinor space tensored with the  $N_t$ -mode Fourier truncation of  $L^2(S_T^1)$ , with fundamental symmetry  $J_L = J_L^{\text{spat}} \otimes I_{N_t}$  at the Bizi–Brouder–Besnard [3] classification  $(m, n) = (4, 6)$  in the Peskin–Schroeder chiral basis, and let  $D_L = i(\gamma^0 \otimes \partial_t + D_{\text{GV}} \otimes I_{N_t})$  be the Lorentzian Dirac per van den Dungen 2016 [32] Proposition 4.1, with  $\partial_t$  Fourier-diagonal anti-Hermitian. Let  $\mathcal{O}_{n_{\max}, N_t, T}^L$  be the truncated Lorentzian operator system built from chirality-doubled scalar 3- $Y$  spatial multipliers tensored with the momentum-diagonal temporal multiplier subalgebra. On the Krein-positive subspace  $\mathcal{K}^+ := \{|\psi\rangle : J_L|\psi\rangle = +|\psi\rangle\}$  we obtain a standard metric spectral triple  $\mathcal{T}_{n_{\max}, N_t, T}^+$  to which Latrémolière 2017/2023 [16] machinery applies verbatim, and define the  $\mathcal{K}^+$ -restricted weak-form Lorentzian propinquity  $\Lambda^L(\mathcal{T}_1^L, \mathcal{T}_2^L) := \Lambda(\mathcal{T}_1^+, \mathcal{T}_2^+)$ . The main theorem reads

$$\Lambda^L(\mathcal{T}_{n_{\max}, N_t, T}^L, \mathcal{T}_{S^3 \times S_T^1}^L) \leq C_3^{\text{joint}} \cdot \gamma_{n_{\max}, N_t, T}^{\text{joint}} \xrightarrow{(n_{\max}, N_t) \rightarrow (\infty, \infty)} 0,$$

with  $C_3^{\text{joint}} \leq 1$  asymptotically tight (inherited from Paper 38 [36] Lemma L3 via a structural identity for the joint commutator), and  $\gamma_{n_{\max}, N_t, T}^{\text{joint}} = O(\log n_{\max}/n_{\max} + T/N_t)$  the joint mass-concentration moment ( $SU(2)$  factor inherits the  $4/\pi$  asymptote of Paper 38 Lemma L2;  $U(1)$  factor inherits the standard Fejér-on-the-circle rate). The proof is a five-lemma argument L1', L2, L3, L4, L5 transferred from Paper 38 with one substantive structural simplification: in the momentum-polynomial convention for temporal multipliers, the joint commutator  $[D_L, a_s \otimes a_t]$  reduces *bit-exactly* to  $i[D_{\text{GV}}, a_s] \otimes a_t$ , so the temporal direction contributes nothing to the joint Lipschitz comparison; the joint  $C_3^{\text{joint}}$  equals the spatial  $C_3^{SU(2)}$  verbatim. The Riemannian limit at  $N_t = 1$  reduces  $\Lambda^L$  to Paper 38's bound bit-exactly (load-bearing falsifier). To our knowledge this is the first Lorentzian propinquity convergence theorem on truncated Krein spectral triples in the published math.OA literature. Honest scope: the construction is a *weak-form* propinquity on the  $\mathcal{K}^+$  Hilbert-space restriction; a *strong-form* Latrémolière-style metric on Krein-signature spectral triples without  $\mathcal{K}^+$  restriction remains an open NCG-math problem. The adjacent synthetic Lorentzian Gromov–Hausdorff program of Mondino–Sämman [20] lives on a categorically different mathematical object (pre-length spaces with causal diamonds, not operator-algebraic spectral triples).

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**Keywords:** Latrémolière propinquity, quantum Gromov–Hausdorff convergence, Krein spectral triple, Bizi–Brouder–Besnard indefinite signature, Camporesi–Higuchi Dirac, spectral truncation, central spectral Fejér kernel, Berezin reconstruction, weak-form propinquity, Krein-positive state space.

## 1 Introduction

This paper closes the convergence-theorem leg of the Lorentzian-extension program for truncated GeoVac spectral triples, following the math.OA quartet Papers 38–40 [36–38] (Riemannian propinquity convergence on  $SU(2)$ , on tensor products of  $SU(2)$  triples at distinct focal lengths, and on general compact Lie groups with bi-invariant metric) and the Lorentzian extensions Papers 42–44 [39–41] (Tomita–Takesaki modular structure on Riemannian truncations; four-witness Wick-rotation at the operator-system level on the Lorentzian Krein wedge; operator-system substrate of the Lorentzian Krein spectral triple at finite cutoff).

The dominant blocker for transferring Latrémolière propinquity [16, 17] to the Lorentzian setting is the absence of a published metric on Krein-signature spectral triples in their own right: Latrémolière’s construction crucially uses the operator-norm Lipschitz seminorm via the Dirac commutator  $\|[D, \cdot]\|_{\text{op}}$ , well-defined on Hilbert space but requiring an indefinite-inner-product analog on a Krein space (with the appropriate Krein-self-adjointness of  $D_L$  per van den Dungen 2016 [32] / BBB [3]). We sidestep this gap by working on the Krein-positive subspace  $\mathcal{K}^+$ , where the Krein product reduces to a positive-definite Hilbert inner product, and the standard Latrémolière machinery applies verbatim. We call this the  *$\mathcal{K}^+$ -restricted weak-form Lorentzian propinquity*.

### 1.1 What this paper does

We work on the compact-temporal Wick-rotated geometry  $S^3 \times S_T^1$ , with Riemannian metric  $ds^2 = d\chi^2 + \sin^2 \chi d\Omega^2 + d\tau^2$  at canonical circumference  $T = 2\pi$ . The Krein structure  $J_L$  is preserved as an *algebraic* object (operator system plus symmetry, in the BBB (4, 6) classification) but no Lorentzian-causal-structure claim is made; the closed timelike-curve obstruction (Geroch [11]) is acknowledged explicitly via the Wick-rotated convention. The construction proceeds as a five-lemma argument transferred from Paper 38:

- (L1′) The Lorentzian operator-system substrate  $\mathcal{O}_{n_{\max}, N_t, T}^L$  of Paper 44 [41], with propagation number 2 on the achievable envelope and trivial Krein-positivity at the operator-multiplier level.
- (L2) A joint central spectral Fejér kernel  $K^{\text{joint}} = K_{n_{\max}}^{\text{SU}(2)} \otimes K_{N_t, T}^{U(1)}$  on  $SU(2) \times U(1)$  with factorized Plancherel symbol and joint cb-norm  $\|S_{K^{\text{joint}}}\|_{\text{cb}} = 2/(n_{\max} + 1)$ , via the Bożejko–Fendler central-multiplier equality [4, 26] on the amenable compact group product.
- (L3) The joint Lichnerowicz / Lipschitz comparison  $\|[D_L, a]\|_{\text{op}} \leq C_3^{\text{joint}} \|\nabla^{\text{joint}} a\|_{\infty}$  with  $C_3^{\text{joint}} \leq 1$ , via the structural identity  $[D_L, a_s \otimes a_t] = i[D_{\text{GV}}, a_s] \otimes a_t$  which makes the temporal direction contribute nothing to the joint commutator.
- (L4) A joint Berezin reconstruction  $B_{n_{\max}, N_t, T}^{\text{joint}} : C^\infty(S^3 \times S_T^1) \rightarrow \mathcal{O}_{n_{\max}, N_t, T}^L$  with positivity, contractivity, approximate identity (rate  $\gamma^{\text{joint}} = O(\log n_{\max}/n_{\max} + T/N_t)$ ), and L3 compatibility.
- (L5) Assembly of the propinquity tunneling pair  $(B^{\text{joint}}, P^{\text{joint}})$  in the Latrémolière weak-form framework on  $\mathcal{K}^+$ , giving the main bound  $\Lambda^L \leq C_3^{\text{joint}} \cdot \gamma^{\text{joint}} \rightarrow 0$ .

## 1.2 Honest scope and named gaps

The construction is at the  $\mathcal{K}^+$ -**restricted weak-form** level. The four named gaps to the strong-form construction are:

- (G1) *Strong-form Lorentzian propinquity*. A Latrémolière-style metric on Krein-signature spectral triples without  $\mathcal{K}^+$  restriction is not in the published literature as of May 2026. The required ingredient is an indefinite-inner-product analog of the operator-norm Lipschitz seminorm with appropriate behavior under Krein-self-adjoint  $D_L$ . This is a multi-month NCG-math problem not addressed here.
- (G2) *De-compactification limit  $T \rightarrow \infty$* . Our construction is at compact temporal radius  $T$  canonical  $T = 2\pi$  (the BW modular period). The non-compact  $\mathbb{R}_t$  limit re-introduces the joint compact  $\times$  non-compact issue. **The norm-resolvent / spectral level of this limit is closed in Paper 47 [43] § 4–5 via a two-rate hybrid composite: the inner arrow is parametric propinquity stability of the present construction at coupled cells  $(n_{\max}, N_t, T(N_t))$  along an admissible scaling  $T(N_t) \rightarrow \infty$  with  $T(N_t)/N_t \rightarrow 0$ , and the outer arrow is norm-resolvent convergence of the Lorentzian Dirac on  $S^3 \times S^1_T$  to the Lorentzian Dirac on  $S^3 \times \mathbb{R}_t$  with explicit exponential tail  $O(e^{-|\operatorname{Im} z| T/2})$  on compact-support subspaces.** The *metric / propinquity* level of this limit on the non-compact carrier remains open (it requires a non-compact extension of Latrémolière’s propinquity, which is not in the published literature as of May 2026): named G2-metric or equivalently G1-genuine on the non-compact substrate (Sprint L3c- $\beta$  / L3e in our internal taxonomy).
- (G3) *Cross-manifold extensions* of the form  $\mathcal{T}_{S^3} \otimes \mathcal{T}_{\text{Hardy}(S^5)}$  are blocked at the NCG-framework level by the Riemannian-vs-Hardy-sector asymmetry (four-layer Coulomb/HO asymmetry from Paper 24 [34] § V) and are not addressed.
- (G4) *Inner-factor calibration data* (the Higgs / Yukawa selection question of the Connes–Marcolli SM construction) is orthogonal to the present convergence theorem.

## 1.3 Related work

Latrémolière’s quantum Gromov–Hausdorff propinquity [17] (Trans. AMS 2018) and the metric-spectral-triple weak-form extension [16] (Adv. Math. 2023) supply the convergence framework we work within. The most recent Riemannian-side continuity result is Latrémolière 2025 [9], treating analytic paths of Riemannian metrics; none of these constructions extend to Krein signature.

Connes–van Suijlekom 2021 [6] establishes the Gromov–Hausdorff convergence of spectral truncations on the circle  $S^1$  and explicitly defers convergence on other manifolds (“elsewhere” on three occasions). The subsequent flat-structure program [8, 13, 14, 18, 30] covers  $\mathbb{T}^d$ , polynomial-growth groups, and the Berezin–Toeplitz  $\mathbb{S}^2$ , with universal  $4/\pi$  asymptote on compact Lie groups established in Paper 40 [38] (also adapted to  $\text{SU}(2)$  in Paper 38 [36]). All of these are strictly Riemannian.

The Lorentzian Krein-spectral-triple framework derives from Strohmaier [28] (axiomatic foundation), Franco–Eckstein [10] (algebraic causality), van den Dungen 2016 [32] (Krein spectral triples and fermionic action; the Wick-rotation lift used here), Bizi–Brouder–Besnard 2018 [3] ( $(m, n)$  classification of indefinite signatures used at (4, 6) West-coast in the present paper), Devastato–Lizzi–Martinetti [7], and the recent  $\text{SU}(1, 1)$  construction [33]; none of these works define a metric on the algebra of operator systems extracted from a Krein triple. The closest algebraic alternative is the Nieuviarts twist morphism [22, 23], which produces pseudo-Riemannian spectral triples from Riemannian ones via a twist on twisted spectral triples; this is not a metric

and moreover (as noted in Paper 42 § 3.2) is restricted to even-dimensional manifolds, so does not apply to  $S^3$  at KO-dimension 3.

The adjacent synthetic Lorentzian Gromov–Hausdorff program of Mondino–Sämman [20] extends their earlier work [19] to Lorentzian pre-length spaces via  $\epsilon$ -nets of causal diamonds; a downstream pure-metric extension via the intrinsic timed-Hausdorff distance appears in Che–Perales–Sormani [21]. This is conceptually adjacent (all are “Lorentzian Gromov–Hausdorff-type convergence”) but lives on a categorically different mathematical object: pre-length spaces with causal-diamond / timed-Hausdorff geometry, not operator-algebraic spectral triples. The two families of constructions are not in competition; neither implies the other.

## 1.4 Outline

§ 2 fixes conventions: Krein space, Lorentzian Dirac, operator system, joint Fejér kernel, Krein-positive subspace, and the definition of the  $\mathcal{K}^+$ -restricted weak-form propinquity. § 3 reviews the  $L1'$  operator-system substrate (Paper 44). § 4 proves the three analytical lemmas L2 (joint cb-norm, § 4.1), L3 (joint Lichnerowicz, § 4.2), and L4 (joint Berezin reconstruction, § 4.3). § 5 states and proves the main theorem (L5 propinquity assembly). § 6 records the numerical verification panel and the bit-exact Riemannian-limit recovery at  $N_t = 1$ . § 7 discusses the four-witness Wick-rotation connection (Paper 42/43), records the open questions (strong-form, de-compactification, cross-manifold), and positions the work against Mondino–Sämman. Appendix A records the symbolic Plancherel calculation for the joint Fejér symbol. Appendix B records the numerical panel data.

## 2 Preliminaries

### 2.1 Compact-temporal manifold and Wick-rotated metric

The compact-temporal Lorentzian / Wick-rotated manifold is  $S^3 \times S_T^1$ , with  $S^3 = \mathrm{SU}(2)$  the round unit three-sphere (spatial slot) and  $S_T^1 = \mathbb{R}/T\mathbb{Z}$  the temporal circle of circumference  $T$ . The Riemannian / Wick-rotated metric is

$$ds^2 = d\chi^2 + \sin^2 \chi d\Omega_2^2 + d\tau^2, \quad (1)$$

with  $\chi \in [0, \pi]$  the polar angle on  $S^3$ ,  $d\Omega_2^2$  the round metric on the  $\mathbb{S}^2$  base of the Hopf fibration, and  $\tau \in \mathbb{R}/T\mathbb{Z}$  the temporal coordinate. We treat the Krein structure  $J_L$  defined below as an *algebraic* object that encodes the BBB (4,6) chirality grading without claiming a Lorentzian-causal interpretation of the underlying metric: Geroch’s no-CTC obstruction [11] prevents a Lorentzian metric on  $S^3 \times S_T^1$  in which  $S_T^1$  is timelike, so the present construction operates on the Wick-rotated geometry and lifts the Krein structure as indefinite-inner-product datum (per Strohmaier [28], vdD [32], BBB [3]).

The canonical choice  $T = 2\pi$  matches the Bisognano–Wichmann [2] modular period at unit surface gravity; this is consistent with the four-witness arc closure of Paper 43 [40]. The proof of the main theorem holds for any  $T > 0$ ; explicit dependence on  $T$  enters the  $U(1)$  factor of the joint mass-concentration moment as  $O(T/N_t)$ .

### 2.2 Krein space and fundamental symmetry

Per Paper 43 [40]§ 2 / Paper 44 [41]§ 2, the Krein space is

$$\mathcal{K}_{n_{\max}, N_t} := \mathcal{H}_{\mathrm{GV}}^{n_{\max}} \otimes \mathbb{C}^{N_t}, \quad (2)$$

where  $\mathcal{H}_{\mathrm{GV}}^{n_{\max}}$  is the chirality-doubled Camporesi–Higuchi [5] spinor Hilbert space truncated at the  $n_{\max}$ -th Fock shell, of dimension

$$\dim \mathcal{H}_{\mathrm{GV}}^{n_{\max}} = \frac{2}{3} n_{\max}(n_{\max} + 1)(n_{\max} + 2), \quad (3)$$

and  $\mathbb{C}^{N_t}$  is the  $N_t$ -mode Fourier truncation of  $L^2(S_T^1)$ . The fundamental symmetry is

$$J_L = J_L^{\text{spat}} \otimes I_{N_t}, \quad (4)$$

with  $J_L^{\text{spat}} = \gamma^0$  the chirality-swap in the Peskin–Schroeder chiral basis,  $(J_L^{\text{spat}})^2 = +I$ .

The BBB  $(m, n)$  classification at signature  $(s, t) = (3, 1)$  West-coast assigns  $(m, n) = (s + t, t - s \bmod 8) = (4, 6)$  via BBB [3] Table 3, fixing  $\epsilon = +1, \epsilon'' = -1, \kappa = -1, \kappa'' = +1$ . These signs were verified bit-exact at finite cutoff in Paper 43 § 5 [40].

### 2.3 Lorentzian Dirac operator

The Lorentzian Dirac, per vdD 2016 [32] Prop. 4.1 applied at signature  $(3, 1)$ , reads

$$D_L = i(\gamma^0 \otimes \partial_t + D_{\text{GV}} \otimes I_{N_t}), \quad (5)$$

with  $\partial_t$  Fourier-diagonal anti-Hermitian on  $\mathbb{C}^{N_t}$ : in the momentum eigenbasis  $\{|k\rangle : k \in \mathbb{Z}, |k| \leq K_{\text{max}}\}$  with  $K_{\text{max}} := \lfloor (N_t - 1)/2 \rfloor$  for odd  $N_t$  or  $N_t/2$  for even  $N_t$ ,

$$\partial_t = i \text{diag}(\omega_k), \quad \omega_k = 2\pi k/T, \quad (6)$$

so  $\partial_t^\dagger = -\partial_t$ . The spatial part  $D_{\text{GV}}$  is the truthful Camporesi–Higuchi spatial Dirac on  $\mathcal{H}_{\text{GV}}^{n_{\text{max}}}$ , diagonal in the spinor basis with eigenvalue  $\chi(n+1/2)$  where  $\chi \in \{+1, -1\}$  is the spinor chirality and  $n$  the Fock-shell index. Krein-self-adjointness  $D_L^\times := J_L D_L^\dagger J_L = D_L$  is verified by direct computation:  $\gamma^{0\dagger} = \gamma^0$  (Hermitian chirality-swap),  $\{\gamma^0, D_{\text{GV}}\} = 0$  in the chiral basis,  $\partial_t^\dagger = -\partial_t$ , and  $J_L^2 = +I$  combine to give the stated identity. The sign  $i^t = +i$  in (5) is forced by this requirement (an opposite sign would give an anti-Krein-self-adjoint  $D_L$ ); see Paper 43 § 3 [40] for the explicit derivation.

### 2.4 Truncated Lorentzian operator system

Per Paper 44 [41], the truncated Lorentzian operator system is the pure-tensor span

$$\mathcal{O}_{n_{\text{max}}, N_t, T}^L := \text{span}_{\mathbb{C}}\{M_{N, L, M}^{\text{spat}} \otimes M_p^{\text{temp}} : (N, L, M) \in \mathcal{I}_{n_{\text{max}}}^{\text{spat}}, p \in \{0, 1, \dots, N_t - 1\}\}, \quad (7)$$

where each spatial multiplier  $M_{N, L, M}^{\text{spat}} = \text{blkdiag}(W_{N, L, M}, W_{N, L, M})$  is the chirality-doubled lift of the Avery–Wen–Avery Weyl-sector multiplier  $W_{N, L, M}$  (the Clebsch–Gordan / Avery 3- $Y$  integral construction of Paper 32 [35] § III), and the temporal multipliers  $M_p^{\text{temp}}$  are momentum-polynomial diagonal:

$$M_p^{\text{temp}} = \text{diag}(\omega_k^p)_{|k| \leq K_{\text{max}}}. \quad (8)$$

Vandermonde invertibility at  $N_t$  distinct frequencies  $\omega_k$  ensures that the span (7) coincides with the span of Fourier-mode indicators  $M_q^{\text{temp}} = \text{diag}(\delta_{k, q})$ , and we use whichever basis is more convenient.

By Paper 44 § 3,  $\mathcal{O}_{n_{\text{max}}, N_t, T}^L$  is a  $*$ -closed linear subspace of  $\mathcal{B}(\mathcal{K}_{n_{\text{max}}, N_t})$ , contains the identity, and is *not* multiplicatively closed (a witness pair is constructed in Paper 44 § 5). The propagation number is  $\text{prop}(\mathcal{O}_{n_{\text{max}}, N_t}^L) = 2$  on the Weyl-doubled achievable envelope, matching the Riemannian Paper 32 § III  $\text{prop}=2$  verbatim;  $\text{prop} = \infty$  on the full  $\mathcal{B}(\mathcal{K})$  envelope due to the chirality-doubling and the commutative temporal subalgebra.

### 2.5 Joint central spectral Fejér kernel

The  $\text{SU}(2)$  factor is the Paper 38 [36] L2-DEF central spectral Fejér kernel

$$K_{n_{\text{max}}}^{\text{SU}(2)}(g) := \frac{1}{Z_{n_{\text{max}}}^{\text{SU}(2)}} \left| \sum_{j \leq j_{\text{max}}} \sqrt{2j+1} \chi_j(g) \right|^2, \quad (9)$$

with  $j_{\max} = (n_{\max} - 1)/2$  and  $Z_{n_{\max}}^{\text{SU}(2)} = n_{\max}(n_{\max} + 1)/2$ , central Plancherel symbol

$$\hat{K}_{n_{\max}}^{\text{SU}(2)}(j) = \frac{2j+1}{Z_{n_{\max}}^{\text{SU}(2)}} \mathbf{1}_{\{j \leq j_{\max}\}}. \quad (10)$$

The  $U(1)$  factor is the standard Fejér kernel on  $S_T^1$ ,

$$K_{N_t, T}^{U(1)}(\theta) := \frac{1}{N_t} \left| \sum_{|k| \leq K_{\max}} e^{i 2\pi k \theta / T} \right|^2, \quad (11)$$

with Plancherel symbol

$$\hat{K}_{N_t}^{U(1)}(k) = \max\left(0, \frac{N_t+1-2|k|}{N_t+1}\right). \quad (12)$$

The joint kernel is the tensor product

$$K_{n_{\max}, N_t, T}^{\text{joint}}(g, \theta) := K_{n_{\max}}^{\text{SU}(2)}(g) \cdot K_{N_t, T}^{U(1)}(\theta), \quad (13)$$

with factorized Plancherel symbol

$$K^{\hat{\text{joint}}}(j, k) = \hat{K}_{n_{\max}}^{\text{SU}(2)}(j) \cdot \hat{K}_{N_t}^{U(1)}(k). \quad (14)$$

Both factors are non-negative, normalized ( $L^1$  norm = 1), and central (class functions on their respective groups); the tensor product inherits these properties on  $\text{SU}(2) \times U(1)$ .

## 2.6 Krein-positive subspace and weak-form propinquity

The Krein-positive subspace is

$$\mathcal{K}^+ := \{|\psi\rangle \in \mathcal{K} : J_L|\psi\rangle = +|\psi\rangle\}, \quad (15)$$

the  $+1$ -eigenspace of  $J_L$ , of dimension  $\dim \mathcal{K}^+ = \dim \mathcal{K}/2 = \frac{1}{3} n_{\max}(n_{\max} + 1)(n_{\max} + 2)N_t$ . The Krein inner product  $\langle \psi, \phi \rangle_{J_L} := \langle \psi, J_L \phi \rangle$  restricted to  $\mathcal{K}^+$  reduces to the standard Hilbert inner product  $\langle \psi, \phi \rangle$ , since  $J_L|\phi\rangle = |\phi\rangle$  on  $\mathcal{K}^+$ . Hence  $\mathcal{K}^+$  is a genuine Hilbert space.

The orthogonal projection onto  $\mathcal{K}^+$  is  $P_+ := (I + J_L)/2$  (and dually  $P_- := (I - J_L)/2$  for the  $-1$ -eigenspace).

**Definition 2.1** ( $\mathcal{K}^+$ -restricted weak-form Lorentzian propinquity). For two Lorentzian truncated Krein metric spectral triples  $\mathcal{T}_i^L = (\mathcal{O}_i^L, \mathcal{K}_i, D_i^L, J_i^L)$  with the construction above, the  $\mathcal{K}^+$ -restricted weak-form Lorentzian propinquity is

$$\Lambda^L(\mathcal{T}_1^L, \mathcal{T}_2^L) := \Lambda(\mathcal{T}_1^+, \mathcal{T}_2^+), \quad (16)$$

where  $\mathcal{T}_i^+ := (P_i^+ \mathcal{O}_i^L P_i^+, \mathcal{K}_i^+, P_i^+ D_i^L P_i^+)$  is the  $\mathcal{K}^+$ -restricted standard metric spectral triple and  $\Lambda$  is the standard Latrémolière metric-spectral-triple propinquity [16].

The construction is well-defined precisely because every Berezin-image operator and every truncation projection used below commutes with  $J_L$  at the operator level (Paper 44 § 4 / Sub-sprint C § 6 of the internal memos), so  $\mathcal{K}^+$  is preserved and the standard machinery applies.

## 2.7 Joint gradient norms

For a pure-tensor symbol  $f = f_s \otimes f_t$  on  $S^3 \times S_T^1$ , the  $L^1$ -additive and  $L^2$ -Pythagorean joint gradient norms are

$$\|\nabla^{\text{joint}, L^1} f\|_{\infty} := \|\nabla_x f_s\|_{\infty} \cdot \|f_t\|_{\infty} + \|f_s\|_{\infty} \cdot \|\partial_t f_t\|_{\infty}, \quad (17)$$

$$\|\nabla^{\text{joint}, L^2} f\|_{\infty} := \sup_{(x, t)} \sqrt{|\nabla_x f(x, t)|^2 + |\partial_t f(x, t)|^2}. \quad (18)$$

For general  $f$ , both extend by linearity over the Peter–Weyl  $\times$  Fourier expansion and obey the standard triangle inequalities. The  $L^2$  form is dominated by the  $L^1$  form via  $\sqrt{a^2 + b^2} \leq a + b$  for  $a, b \geq 0$ ; both forms upper-bound the same Lipschitz seminorm with the same constant in the proof of Lemma 4.4.

### 3 Lemma L1': the operator-system substrate

Lemma L1' supplies the truncated metric spectral triple on which the four subsequent lemmas operate. It is the closure result of Paper 44 [41], summarized here for completeness.

**Lemma 3.1** (L1', Paper 44). *The collection  $\mathcal{T}_{n_{\max}, N_t, T}^L := (\mathcal{O}_{n_{\max}, N_t, T}^L, \mathcal{K}_{n_{\max}, N_t}, D_{n_{\max}, N_t, T}^L, J_{n_{\max}, N_t}^L)$  defined in § 2.2–2.4 is a Lorentzian truncated Krein metric spectral triple in the sense of vdD [32] / BBB [3]:*

- (a)  $\mathcal{O}^L$  is a unital  $*$ -closed linear subspace of  $\mathcal{B}(\mathcal{K})$ .
- (b)  $D_L$  is Krein-self-adjoint,  $D_L^\times = D_L$ .
- (c)  $J_L$  is a fundamental symmetry,  $J_L^2 = +I$ ,  $J_L^\dagger = J_L$ .
- (d) The propagation number  $\text{prop}(\mathcal{O}^L) = 2$  on the Weyl-doubled achievable envelope.
- (e) Every multiplier in  $\mathcal{O}^L$  commutes with  $J_L$ :  $[J_L, a] = 0$  bit-exact for every  $a \in \mathcal{O}^L$ .
- (f) At  $N_t = 1$ , the construction reduces bit-exactly to the Riemannian chirality-doubled truncated spectral triple on  $\mathcal{H}_{\text{GV}}^{n_{\max}}$  (Frobenius residual 0.0 in float64).

*Proof.* (a)–(c) are direct from the construction and Paper 43 § 2,3,5. (d) is Paper 44 § 3 (via the structural identity  $\dim((\mathcal{O}^L)^2) = \dim_{\text{Weyl}}^2 N_t$  on the Weyl-doubled envelope). (e) follows from  $[\gamma^0, M^{\text{spat}}] = 0$  (chirality-swap acts trivially on the diagonal block-doubled spatial multiplier) and  $[I_{N_t}, M^{\text{temp}}] = 0$  (trivially). (f) is Paper 44 § 2 (load-bearing falsifier F1).  $\square$

**Remark 3.2** (Krein-positivity at the operator-multiplier level is trivial). By Lemma 3.1(e), every operator  $a \in \mathcal{O}^L$  preserves both  $\mathcal{K}^+$  and  $\mathcal{K}^-$ . Consequently, the  $\mathcal{K}^+$ -restricted operator system  $\mathcal{O}^{L,+} := P_+ \mathcal{O}^L P_+$  has the same propagation number 2 as the unrestricted system, and the standard Latrémolière metric-spectral-triple machinery applies on  $(\mathcal{O}^{L,+}, \mathcal{K}^+, P_+ D_L P_+)$  verbatim. The non-trivial Krein-positive structure shifts to the *state* level (Wasserstein–Kantorovich on  $\mathcal{K}^+$  states), which is the natural setting for the L5 propinquity assembly in § 5.

## 4 The three analytical lemmas

We now prove the three analytical lemmas L2 (joint cb-norm, § 4.1), L3 (joint Lichnerowicz, § 4.2), and L4 (joint Berezin reconstruction, § 4.3) on the substrate supplied by Lemma 3.1. The mechanisms transfer from Paper 38 / Paper 40 with one substantive simplification per lemma, arising from the pure-tensor structure of  $\mathcal{O}^L$  and the momentum-diagonal structure of the temporal multipliers.

### 4.1 Lemma L2: joint cb-norm of the central Schur multiplier

**Lemma 4.1** (L2, joint cb-norm). *For all  $n_{\max} \geq 1$  and  $N_t \geq 1$ , the central Schur multiplier  $S_{K^{\text{joint}}}$  on the central subalgebra  $Z(C(\text{SU}(2) \times U(1)))$  has cb-norm*

$$\|S_{K^{\text{joint}}}_{n_{\max}, N_t, T}\|_{\text{cb}} = \|S_{K^{\text{SU}(2)}}\|_{\text{cb}} \cdot \|S_{K^{U(1)}}\|_{\text{cb}} = \frac{2}{n_{\max} + 1} \cdot 1 = \frac{2}{n_{\max} + 1}. \quad (19)$$

*Proof.* The compact group  $G = \text{SU}(2) \times U(1)$  is amenable (direct product of two compact groups, both amenable by Haar invariance [1, 25]). By the Bożejko–Fendler central-multiplier equality [4, 26] Ch. 8 Thm. 8.10, every central Fourier multiplier symbol  $m \in \ell^\infty(\hat{G})$  has Schur cb-norm equal to its  $\ell^\infty$  norm.

*Factor 1* ( $\text{SU}(2)$ ). Inherited from Paper 38 [36] Lemma L2(c) verbatim: the Plancherel symbol  $\hat{K}_{n_{\max}}^{\text{SU}(2)}(j) = (2j+1)/Z_{n_{\max}}^{\text{SU}(2)}$  is monotonically increasing in  $j \in \{0, \frac{1}{2}, 1, \dots, j_{\max}\}$ , so its supremum is  $\hat{K}_{n_{\max}}^{\text{SU}(2)}(j_{\max}) = n_{\max}/Z_{n_{\max}}^{\text{SU}(2)} = 2/(n_{\max}+1)$ .

*Factor 2* ( $U(1)$ ).  $U(1)$  is abelian, so all irreps are 1-dimensional and every multiplier symbol is automatically central. The Fejér symbol (12) is non-negative, supported on  $|k| \leq K_{\max}$ , and monotonically decreasing in  $|k|$  from  $\hat{K}_{N_t}^{U(1)}(0) = 1$ , so its supremum is 1, attained at  $k = 0$ .

*Joint cb-norm via factorization.* The central subalgebra of  $C(\text{SU}(2) \times U(1))$  factorizes as the spatial  $C^*$ -tensor product  $Z(C(\text{SU}(2))) \hat{\otimes} Z(C(U(1)))$  (Takesaki [29] Thm. IV.4.14, commutative-factor case). The joint Plancherel symbol (14) factorizes correspondingly. Both factor symbols are non-negative, so

$$\sup_{(j,k)} \hat{K}^{\text{SU}(2)}(j) \hat{K}^{U(1)}(k) = \left( \sup_j \hat{K}^{\text{SU}(2)}(j) \right) \left( \sup_k \hat{K}^{U(1)}(k) \right).$$

Applying Bożejko–Fendler in reverse to each factor (and once jointly on the amenable product) gives (19).  $\square$

**Remark 4.2** ( $N_t$ -independence of the cb-norm). The joint cb-norm  $2/(n_{\max}+1)$  is independent of  $N_t$ : the  $U(1)$  factor caps at 1 regardless of  $N_t$ . The  $N_t$ -dependence of the joint convergence rate enters the *mass-concentration moment*  $\gamma^{\text{joint}}$  in Lemma 4.9, not the cb-norm. This is the structurally correct behavior: the cb-norm controls the Lipschitz-distortion height in the propinquity assembly (§ 5), while  $\gamma^{\text{joint}}$  controls the reach. They are different invariants of the same kernel.

**Remark 4.3** ( $4/\pi$  asymptote on the  $\text{SU}(2)$  factor). Paper 38 § 3.2 (with the universal Plancherel- $\times$ -Vandermonde cancellation theorem from Paper 40 § 3.2 [38]) gives the asymptotic rate constant

$$\lim_{n_{\max} \rightarrow \infty} \frac{n_{\max} \cdot \gamma_{n_{\max}}^{\text{SU}(2)}}{\log n_{\max}} = \frac{4}{\pi}.$$

This is the universal compact-Lie-group rate constant on the bi-invariant metric and is independent of rank, Weyl-group order, and Lie type within the class. The  $U(1)$  factor inherits a separate Fejér-on-the-circle rate.

## 4.2 Lemma L3: joint Lichnerowicz / Lipschitz comparison

**Lemma 4.4** (L3, joint Lichnerowicz). *For every  $a \in \mathcal{O}_{n_{\max}, N_t, T}^L$  and every  $(n_{\max}, N_t, T)$ ,*

$$\|[D_L, a]\|_{\text{op}} \leq C_3^{\text{joint}}(n_{\max}, N_t) \cdot \|\nabla^{\text{joint}} a\|_{\infty}, \quad (20)$$

where  $\nabla^{\text{joint}}$  is either the  $L^1$ -additive or  $L^2$ -Pythagorean joint gradient norm (17), (18). The constant satisfies

$$C_3^{\text{joint}}(n_{\max}, N_t) \leq C_3^{\text{SU}(2)}(n_{\max}) \leq \sup_{2 \leq N \leq 2n_{\max}-1} \frac{N-1}{\sqrt{N^2-1}} = \sqrt{1 - \frac{1}{n_{\max}}} \xrightarrow{n_{\max} \rightarrow \infty} 1^-, \quad (21)$$

where the supremum is taken over the natural-substrate envelope  $N \leq 2n_{\max}-1$  — the Avery–Wen–Avery  $\Delta n = \pm 1$  ladder on  $n_{\max}$  shells produces shell-difference labels  $N$  up to  $2n_{\max}-1$  via the chirality-doubling at the operator-system level (see Paper 44 [41] § 3). The per-harmonic function  $(N-1)/\sqrt{N^2-1}$  is Paper 38 [36] Lemma L3 verbatim; the envelope range  $N \leq 2n_{\max}-1$  extends Paper 38’s  $N \leq n_{\max}$  because Paper 38’s Berezin map restricts to multiplier labels  $N \leq n_{\max}$  by Plancherel cutoff, while the Lorentzian natural substrate’s full multiplier algebra reaches the larger envelope (see Remark 4.5 below).



**Remark 4.5** (On the supremum range). The supremum is taken over the natural-substrate envelope  $N \leq 2n_{\max} - 1$  rather than  $N \leq n_{\max}$ . The per-harmonic functional form, the asymptotic limit  $1^-$ , and the main theorem (Theorem 5.3) are insensitive to this distinction. The finite-cutoff constant is sharpened by the envelope: at  $n_{\max} = 3$ , the natural-substrate generator  $(N, L, M, p) = (4, 1, \pm 1, 0)$  has Lipschitz ratio 0.95, which exceeds the per-shell value  $C_3^{(3)} = 2/\sqrt{8} = 0.707$  but is bounded by the envelope-aware  $C_3^{\text{op}}(3) = \sqrt{(2 \cdot 3 - 1) - 1}/\sqrt{(2 \cdot 3 - 1)^2 - 1} = \sqrt{2/3} \approx 0.816$ . Numerical-panel values in §6.1 are unaffected (they quote  $\gamma^{\text{joint}}$  directly, not  $C_3^{\text{joint}} \cdot \gamma^{\text{joint}}$ ). The envelope refinement was identified in Paper 46 [42] § 4.3 (the strong-form companion under  $L_{\text{op}}(a) := \|[D_L, a]\|_{\text{op}}$ ); Paper 38 is unaffected because Paper 38’s Berezin map restricts multiplier labels to  $N \leq n_{\max}$  by Plancherel cutoff, while the Lorentzian natural substrate’s chirality-doubled multiplier algebra reaches the larger envelope (see Paper 46 Appendix A for the cross-paper analysis).

*Proof.* We first establish the structural identity: for every pure-tensor  $a = a_s \otimes a_t \in \mathcal{O}^L$ ,

$$[D_L, a_s \otimes a_t] = i [D_{\text{GV}}, a_s] \otimes a_t \quad (\text{bit-exact}). \quad (22)$$

Expand  $[D_L, a_s \otimes a_t] = i [\gamma^0 \otimes \partial_t, a_s \otimes a_t] + i [D_{\text{GV}} \otimes I_{N_t}, a_s \otimes a_t]$  and analyze each term.

*Term A (vanishing cross term).* Since  $a_s = \text{blkdiag}(W, W)$  has identical diagonal blocks in chirality and  $\gamma^0$  is the chirality-swap acting as  $\begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}$  on the chirality indices,

$$\gamma^0 a_s = \begin{pmatrix} 0 & W \\ W & 0 \end{pmatrix} = a_s \gamma^0, \quad [\gamma^0, a_s] = 0.$$

Furthermore,  $a_t$  and  $\partial_t$  are both diagonal in the momentum basis (the temporal multiplier algebra is the commutative diagonal subalgebra of  $M_{N_t}(\mathbb{C})$ ), so  $[\partial_t, a_t] = 0$ . The cross term therefore vanishes bit-exactly:  $[\gamma^0 \otimes \partial_t, a_s \otimes a_t] = 0$ .

*Term B (purely spatial).*  $[D_{\text{GV}} \otimes I_{N_t}, a_s \otimes a_t] = [D_{\text{GV}}, a_s] \otimes a_t$ .

Combining gives (22). By linearity over the operator-system span, the same identity extends to every  $a \in \mathcal{O}^L$  written as a finite linear combination of pure-tensor generators.

Tensor-product operator-norm factorization  $\|A \otimes B\|_{\text{op}} = \|A\|_{\text{op}} \cdot \|B\|_{\text{op}}$  and Paper 38 Lemma L3 on the spatial factor give, for any pure-tensor  $a = a_s \otimes a_t$ ,

$$\|[D_L, a]\|_{\text{op}} = \|[D_{\text{GV}}, a_s]\|_{\text{op}} \cdot \|a_t\|_{\text{op}} \leq C_3^{\text{SU}(2)}(n_{\max}) \cdot \|\nabla_x f_s\|_{\infty} \cdot \|a_t\|_{\text{op}}, \quad (23)$$

where  $f_s$  is the Berezin pre-image of  $a_s$ . For momentum-diagonal  $a_t$ ,  $\|a_t\|_{\text{op}} = \|f_t\|_{\infty}$  (with  $f_t$  the corresponding Fourier symbol). Since  $\|\nabla_x f_s\|_{\infty} \cdot \|f_t\|_{\infty} \leq \|\nabla^{\text{joint}, L^1} f\|_{\infty}$  by (17) (the second summand is non-negative), and similarly  $\leq \|\nabla^{\text{joint}, L^2} f\|_{\infty}$  by  $\sqrt{a^2 + b^2} \geq a$  for  $a, b \geq 0$ , we obtain (20) for pure-tensor  $a$ . The extension to general  $a = \sum_j c_j (a_s^j \otimes a_t^j)$  follows by triangle inequality on both sides.

The bound (21) takes the Paper 38 § L3 per-harmonic form  $(N-1)/\sqrt{N^2-1}$  on the natural-substrate envelope  $N \leq 2n_{\max} - 1$  (per Remark 4.5); asymptotic tightness  $\rightarrow 1^-$  is via the per-harmonic Avery–Wen–Avery saturation on  $Y_{N,0,0}^{(3)}$  as  $N \rightarrow 2n_{\max} - 1$ .  $\square$

**Remark 4.6** (The temporal direction contributes zero). The identity (22) is the substantive structural content of Lemma 4.4. There is *no* cross term and *no* separate  $U(1)$  Lichnerowicz lemma: the joint commutator is the spatial commutator times the temporal operator-norm, and the temporal slot enters the bound multiplicatively at value 1. This is stronger than the  $C_3 \leq 1 + \text{cross estimate}$  one would expect from a generic tensor extension; the structural reason is that the temporal multiplier algebra is commutative diagonal and  $\partial_t$  is itself diagonal, so they commute bit-exactly. The Wick-rotated convention here is essential: a Lorentzian-causal temporal derivative would not be diagonal in the same basis.

**Remark 4.7** ( $L^1$  and  $L^2$  give the same  $C_3$ ). Both  $L^1$ -additive (17) and  $L^2$ -Pythagorean (18) joint gradient norms upper-bound the right-hand side of (23) by the same constant  $C_3^{\text{SU}(2)}$ : the temporal contribution to  $\| [D_L, a] \|_{\text{op}}$  is zero (by (22)), so only the spatial part of the joint gradient matters in the inequality.

### 4.3 Lemma L4: joint Berezin reconstruction

**Definition 4.8** (Joint Berezin reconstruction). The joint Berezin reconstruction map is  $B_{n_{\max}, N_t, T}^{\text{joint}} : C^\infty(S^3 \times S_T^1) \rightarrow \mathcal{O}_{n_{\max}, N_t, T}^L$  defined on the joint Peter–Weyl  $\times$  Fourier expansion  $f(\omega, t) = \sum_{N, L, M, q} c_{N, L, M, q} Y_{N, L, M}^{(3)}(\omega) e^{iqt/R_T}$  (with  $R_T = T/(2\pi)$ ) by

$$B^{\text{joint}}(f) = \sum_{\substack{N \leq n_{\max} \\ |q| \leq K_{\max}}} \sum_{L=0}^{N-1} \sum_{|M| \leq L} \hat{K}_{n_{\max}}^{\text{SU}(2)}(N) \hat{K}_{N_t}^{U(1)}(q) c_{N, L, M, q} (M_{N, L, M}^{\text{spat}} \otimes M_q^{\text{temp}}). \quad (24)$$

Equivalently,  $B^{\text{joint}}(f) = P^{\text{joint}} (K^{\text{joint}} * f) P^{\text{joint}}$  where  $P^{\text{joint}} := P_{n_{\max}}^{\text{SU}(2)} \otimes P_{N_t}^{U(1)}$  is the joint truncation projection onto the truncated subspace of  $L^2(S^3 \times S_T^1)$  spanned by the kept Peter–Weyl  $\times$  Fourier modes. The two definitions agree on the central subalgebra by joint Plancherel (14).

**Lemma 4.9** (L4, joint Berezin properties).  $B_{n_{\max}, N_t, T}^{\text{joint}}$  satisfies the following four properties.

- (a) Positivity. If  $f \geq 0$  pointwise on  $S^3 \times S_T^1$ , then  $B^{\text{joint}}(f) \geq 0$  as a positive-semidefinite Hermitian element of  $M_{\dim \mathcal{K}}(\mathbb{C})$ .
- (b) Contractivity.  $\|B^{\text{joint}}\|_{\text{cb}} \leq \|S_{K^{\text{joint}}}\|_{\text{cb}} = 2/(n_{\max}+1) \leq 1$ , and in particular  $\|B^{\text{joint}}(f)\|_{\text{op}} \leq \|f\|_\infty$ .
- (c) Approximate identity. For every  $f \in C^\infty(S^3 \times S_T^1)$ ,

$$\|B^{\text{joint}}(f) - P^{\text{joint}} M_f P^{\text{joint}}\|_{\text{op}} \leq \gamma_{n_{\max}, N_t, T}^{\text{joint}} \cdot \|\nabla^{\text{joint}} f\|_\infty, \quad (25)$$

with  $\gamma^{\text{joint}} = O(\log n_{\max}/n_{\max} + T/N_t) \rightarrow 0$  as  $(n_{\max}, N_t) \rightarrow (\infty, \infty)$ .

- (d) L3 compatibility. For every  $f \in C^\infty(S^3 \times S_T^1)$ ,

$$\|[D_L, B^{\text{joint}}(f)]\|_{\text{op}} \leq C_3^{\text{joint}} \cdot \|\nabla^{\text{joint}} f\|_\infty, \quad (26)$$

with the same  $C_3^{\text{joint}} \leq 1$  as in Lemma 4.4.

- (e) Krein-positivity preservation. For every  $f$ ,  $[J_L, B^{\text{joint}}(f)] = 0$  bit-exact, hence  $B^{\text{joint}}(f)$  preserves both  $\mathcal{K}^+$  and  $\mathcal{K}^-$ .

*Proof.* (a) *Positivity.* The convolution form  $B^{\text{joint}}(f) = P^{\text{joint}} (K^{\text{joint}} * f) P^{\text{joint}}$  writes  $B^{\text{joint}}(f)$  as a UCP compression of a non-negative function (since both factor kernels are non-negative pointwise:  $K^{\text{SU}(2)} = |\cdot|^2 \cdot Z^{-1} \geq 0$  and  $K^{U(1)} = |\cdot|^2 \cdot N_t^{-1} \geq 0$ ). The convolution of two non-negative integrands against the positive joint Haar measure is non-negative, and the UCP compression of a non-negative multiplication operator is positive-semidefinite.

(b) *Contractivity.* The convolution form gives  $\|B^{\text{joint}}(f)\|_{\text{op}} \leq \|K^{\text{joint}} * f\|_\infty \leq \|K^{\text{joint}}\|_{L^1} \cdot \|f\|_\infty$  by Young’s inequality at the sup endpoint, and  $\|K^{\text{joint}}\|_{L^1} = 1$  (both factors are normalized probability densities; Haar on a product factorizes). The sharper cb-norm bound is via Lemma 4.1:  $B^{\text{joint}}$  is the composition of the UCP projection  $P^{\text{joint}}$  (cb-norm 1) with the central Schur multiplier  $S_{K^{\text{joint}}}$  (cb-norm  $2/(n_{\max}+1)$  by Lemma 4.1), giving  $\|B^{\text{joint}}\|_{\text{cb}} \leq 2/(n_{\max}+1)$ .

(c) *Approximate identity.* For a pure-tensor symbol  $f = f_s \otimes f_t$ , the convolution-deficit splits:

$$K^{\text{joint}} * f - f = (K^{\text{SU}(2)} * f_s - f_s) \otimes (K^{U(1)} * f_t) + f_s \otimes (K^{U(1)} * f_t - f_t).$$

By Paper 38 [36] Lemma L4(c) on the spatial factor,  $\|K^{\text{SU}(2)} * f_s - f_s\|_\infty \leq \gamma_{n_{\max}}^{\text{SU}(2)} \cdot \|\nabla_x f_s\|_\infty$ , with  $\gamma_{n_{\max}}^{\text{SU}(2)} = O(\log n_{\max}/n_{\max})$  (Paper 38 § L2 quantitative rate,  $4/\pi$  asymptote). By the standard Fejér-on-the-circle approximation rate (Katznelson [15] Ch. I),  $\|K^{U(1)} * f_t - f_t\|_\infty \leq \gamma_{N_t, T}^{U(1)} \cdot \|\partial_t f_t\|_\infty$  with  $\gamma^{U(1)} = O(T/N_t)$ . Combining with the  $L^1$ -additive triangle decomposition and  $\|K^{U(1)} * f_t\|_\infty \leq \|f_t\|_\infty$  (Young) gives

$$\|K^{\text{joint}} * f - f\|_\infty \leq \max(\gamma_{n_{\max}}^{\text{SU}(2)}, \gamma_{N_t, T}^{U(1)}) \cdot \|\nabla^{\text{joint}, L^1} f\|_\infty.$$

We define  $\gamma^{\text{joint}} := \max(\gamma^{\text{SU}(2)}, \gamma^{U(1)})$ . Compression-contraction on  $L^2$  does not increase the operator norm, so (25) follows. Linearity and the triangle inequality extend to general  $f$ .

(d) *L3 compatibility.* By linearity over (24),  $[D_L, B^{\text{joint}}(f)] = \sum c_{N, L, M, q} K^{\hat{\text{joint}}}(N, q) i[D_{\text{GV}}, M_{N, L, M}^{\text{spat}}] \otimes M_q^{\text{temp}}$  by the L3 structural identity (22). Tensor-norm factorization and Paper 38 § L3 give  $\|[D_{\text{GV}}, M_{N, L, M}^{\text{spat}}]\|_{\text{op}} \leq (N-1) \cdot \|M^{\text{spat}}\|_{\text{op}}$  with  $C_3^{(N)} = (N-1)/\sqrt{N^2-1}$  on the natural Avery harmonic. Combining with  $K^{\hat{\text{joint}}}(N, q) \leq 1$  and triangle inequality gives (26).

(e) *Krein-positivity preservation.*  $[J_L, M_{N, L, M}^{\text{spat}} \otimes M_q^{\text{temp}}] = [J_L^{\text{spat}}, M^{\text{spat}}] \otimes M^{\text{temp}} + M^{\text{spat}} \otimes [I_{N_t}, M^{\text{temp}}] = 0$  by the same argument as Lemma 3.1(e), extended by linearity.  $\square$

**Remark 4.10** (Tensor-product factorization on pure tensors). For a pure-tensor  $f = f_s \otimes f_t$ , the joint Berezin factorizes:

$$B^{\text{joint}}(f_s \otimes f_t) = B_{\chi\text{-d}}^{\text{SU}(2)}(f_s) \otimes B^{U(1)}(f_t), \quad (27)$$

where  $B_{\chi\text{-d}}^{\text{SU}(2)}$  is the chirality-doubled spinor lift of Paper 38's spatial Berezin and  $B^{U(1)}$  is the standard Fejér-Berezin on  $S_T^1$ . This factorization makes Lemma 4.9(a)–(e) follow from the corresponding factor-wise properties of Paper 38 L4 (SU(2) side) and standard Fejér-on- $S^1$  approximation theory ( $U(1)$  side).

## 5 Main theorem (Lemma L5): propinquity assembly

### 5.1 The tunneling pair

**Definition 5.1** (Joint tunneling pair). The joint tunneling pair between the truncated triple  $\mathcal{T}_{n_{\max}, N_t, T}^L$  and the continuum reference triple  $\mathcal{T}_{S^3 \times S_T^1}^L$  is the pair  $(B_{n_{\max}, N_t, T}^{\text{joint}}, P_{n_{\max}, N_t}^{\text{joint}})$ , with  $B^{\text{joint}}$  the joint Berezin map of Definition 4.8 and  $P^{\text{joint}} = P_{n_{\max}}^{\text{SU}(2)} \otimes P_{N_t}^{U(1)}$  the joint truncation projection on  $L^2(S^3 \times S_T^1)$ .

The continuum reference triple is  $\mathcal{T}_{S^3 \times S_T^1}^L := (C^\infty(S^3 \times S_T^1), L^2(S^3, \Sigma_{\text{CH}}) \otimes L^2(S_T^1), D_\infty^L, J_\infty^L)$  with  $\Sigma_{\text{CH}}$  the chirality-doubled Camporesi–Higuchi spinor bundle and  $D_\infty^L, J_\infty^L$  the natural continuum analogs of  $D_L$  and  $J_L$ .

By Lemmas 3.1–4.9, both  $B^{\text{joint}}$  and  $P^{\text{joint}}$  are UCP (with the Berezin cb-norm bound  $\|B^{\text{joint}}\|_{\text{cb}} \leq 2/(n_{\max} + 1) \leq 1$ ; the projection  $P^{\text{joint}}$  is a Stinespring orthogonal projection), and both commute with  $J_L$  at the operator level. Hence the  $\mathcal{K}^+$ -restricted pair  $(P_+ B^{\text{joint}}(\cdot) P_+, P_+ P^{\text{joint}}(\cdot) P_+)$  is a valid Latrémolière tunneling pair between the  $\mathcal{K}^+$ -restricted truncated and continuum triples in Definition 2.1.

## 5.2 The four propinquity constituents

Following Latrémolière [16]§ 4 (metric-spectral-triple weak-form propinquity bound) and Paper 38 § L5, the propinquity bound for a direct UCP tunneling pair is

$$\Lambda(\mathcal{T}_{n_{\max}, N_t, T}^+, \mathcal{T}_{S^3 \times S_T^1}^+) \leq \max(\text{reach}_B, \text{reach}_P, \text{height}_B, \text{height}_P). \quad (28)$$

**Proposition 5.2** (Reach and height bounds). *The four propinquity constituents satisfy:*

$$\text{reach}_B \leq \gamma_{n_{\max}, N_t, T}^{\text{joint}}, \quad (29)$$

$$\text{reach}_P \leq \gamma_{n_{\max}, N_t, T}^{\text{joint}}, \quad (30)$$

$$\text{height}_B \leq C_3^{\text{joint}} \cdot \gamma_{n_{\max}, N_t, T}^{\text{joint}}, \quad (31)$$

$$\text{height}_P = 0. \quad (32)$$

*Proof.* (29) is Lemma 4.9(c) (approximate identity), applied on the unit-Lipschitz ball  $\{f : \|\nabla^{\text{joint}} f\|_\infty \leq 1\}$ . The bound restricts unchanged to  $\mathcal{K}^+$  since  $P_+$  is UCP.

(30) is the dual roundtrip: start with  $M_f$  on the continuum side, apply  $P^{\text{joint}}$  to compress to  $\mathcal{O}^L$ , and lift back with a bounded partial inverse  $\sigma$  of  $B^{\text{joint}}$  on the central subalgebra. By Lemma 4.1,  $\|S_{K^{\text{joint}}}\|_{\text{cb}} = 2/(n_{\max}+1)$  is finite and non-zero, so  $\sigma$  exists on the central subalgebra and is bounded by the same scale; the dual residual is symmetric to (29).

(31) is the Lipschitz-distortion height of Latrémolière [16] Def. 3.5, given by

$$\text{height}_B = \sup_{\|\nabla^{\text{joint}} f\|_\infty \leq 1} \left| \|f\|_{\text{Lip}} - \|B^{\text{joint}}(f)\|_{\text{Lip}^{\text{op}}} \right|,$$

where  $\|B^{\text{joint}}(f)\|_{\text{Lip}^{\text{op}}} := \|[D_L, B^{\text{joint}}(f)]\|_{\text{op}}$  is the Lipschitz seminorm induced by the truncated Lorentzian Dirac. By Lemma 4.9(d) and an L4 approximate-identity step on the gradient (Stein-Weiss / Paper 38 Appendix A factor-wise), the height is bounded by  $C_3^{\text{joint}} \cdot \gamma^{\text{joint}}$ .

(32) follows because  $P^{\text{joint}}$  is an orthogonal projection (Stinespring lift), of cb-norm 1, and introduces no Lipschitz distortion.  $\square$

## 5.3 Main theorem

**Theorem 5.3** (Lorentzian propinquity convergence on truncated  $\text{SU}(2) \times U(1)_T$  Krein spectral triples). *Let  $n_{\max} \geq 1$ ,  $N_t \geq 1$ ,  $T > 0$ . The  $\mathcal{K}^+$ -restricted weak-form Lorentzian Latrémolière propinquity between the truncated Krein metric spectral triple  $\mathcal{T}_{n_{\max}, N_t, T}^L$  and the continuum reference triple  $\mathcal{T}_{S^3 \times S_T^1}^L$  satisfies*

$$\Lambda^L(\mathcal{T}_{n_{\max}, N_t, T}^L, \mathcal{T}_{S^3 \times S_T^1}^L) \leq C_3^{\text{joint}}(n_{\max}, N_t) \cdot \gamma_{n_{\max}, N_t, T}^{\text{joint}} \xrightarrow{(n_{\max}, N_t) \rightarrow (\infty, \infty)} 0, \quad (33)$$

with  $C_3^{\text{joint}}(n_{\max}, N_t) \leq 1$  asymptotically tight (inherited from Paper 38 / Lemma 4.4) and  $\gamma_{n_{\max}, N_t, T}^{\text{joint}} = O(\log n_{\max}/n_{\max} + T/N_t)$  the joint mass-concentration moment (Lemma 4.9(c)).

*Proof.* By Definition 2.1 and Proposition 5.2,

$$\begin{aligned} \Lambda^L(\mathcal{T}_{n_{\max}, N_t, T}^L, \mathcal{T}_{S^3 \times S_T^1}^L) &= \Lambda(\mathcal{T}_{n_{\max}, N_t, T}^+, \mathcal{T}_{S^3 \times S_T^1}^+) \\ &\leq \max(\gamma^{\text{joint}}, \gamma^{\text{joint}}, C_3^{\text{joint}} \cdot \gamma^{\text{joint}}, 0) \\ &= C_3^{\text{joint}} \cdot \gamma_{n_{\max}, N_t, T}^{\text{joint}}. \end{aligned}$$

Since  $\gamma^{\text{joint}} = O(\log n_{\max}/n_{\max} + T/N_t)$  vanishes in the joint limit  $(n_{\max}, N_t) \rightarrow (\infty, \infty)$  and  $C_3^{\text{joint}} \leq 1$  asymptotically, the right-hand side  $\rightarrow 0$ .  $\square$

**Remark 5.4** (Asymptotic rate). The joint rate factors at leading order:

$$\gamma^{\text{joint}} \leq \gamma^{\text{SU}(2)} + \gamma^{U(1)} = O\left(\frac{\log n_{\max}}{n_{\max}}\right) + O\left(\frac{T}{N_t}\right),$$

with the  $\text{SU}(2)$  leading constant  $\lim_{n_{\max} \rightarrow \infty} n_{\max} \gamma^{\text{SU}(2)} / \log n_{\max} = 4/\pi$  rigorous (Paper 38 § L2 Stein–Weiss closure + Paper 40 § 3.2 universal cancellation), and the  $U(1)$  leading constant  $\gamma_{N_t, T}^{U(1)} \leq C T \log N_t / N_t$  standard (Katznelson [15] Ch. I).

**Remark 5.5** (Limit identification). The propinquity limit point is the  $\mathcal{K}^+$ -restricted standard metric spectral triple  $\mathcal{T}_{S^3 \times S_T^1}^+ = (P_+ C^\infty(S^3 \times S_T^1) P_+, \mathcal{K}_\infty^+, P_+ D_\infty^L P_+)$  on the  $\mathcal{K}^+$  subspace of the continuum Krein space. By Latrémolière’s weak-form theorem [16] Thm. 5.5, the limit identification of  $\Lambda$ -convergence is on the Wasserstein–Kantorovich state space of the limit triple; this carries over to  $\Lambda^L$  via Definition 2.1. The weak-form caveat is that the strong-form Lorentzian propinquity (without  $\mathcal{K}^+$  restriction) would identify a Krein-state-space limit, which remains open; see § 7.3.

**Remark 5.6** (The proof shape is bookkeeping). In the same spirit as Paper 38 § L5 and Leimbach–vS [18], the propinquity assembly is the bookkeeping layer of the proof shape. The analytical content is supplied by Lemmas 3.1, 4.1, 4.4, 4.9; each step in Proposition 5.2 cites one of them. No new analytical input enters the assembly.

**Remark 5.7** (Two-rate hybrid convergence to the non-compact carrier). The construction here is at *fixed* compact temporal radius  $T$  (canonical  $T = 2\pi$ ). At any *admissible scaling*  $T(N_t) \rightarrow \infty$  with  $T(N_t)/N_t \rightarrow 0$  as  $N_t \rightarrow \infty$ , the present theorem applies at every coupled cell  $(n_{\max}, N_t, T(N_t))$  with  $C_3^{\text{joint}}(n_{\max}) = \sqrt{1 - 1/n_{\max}}$  ( $T$ -independent by L3 structural identity (22)) and  $\gamma^{\text{joint}} = \max(O(\log n_{\max}/n_{\max}), O(T(N_t)/N_t)) \rightarrow 0$  (the  $U(1)_T$  factor of L4 carries the only  $T$ -dependence in the bound, and it is the standard Fejér-on-circle rate). Composed with the norm-resolvent convergence  $\mathcal{T}_{S^3 \times S_T^1}^L \rightarrow \mathcal{T}_{S^3 \times \mathbb{R}_t}^L$  of Paper 47 [43] § 4, this gives a *two-rate hybrid convergence* statement

$$\mathcal{T}_{n_{\max}, N_t, T(N_t)}^L \xrightarrow{\Lambda^L\text{-rate } O(\log n_{\max}/n_{\max} + T(N_t)/N_t)} \mathcal{T}_{S^3 \times S_{T(N_t)}^1}^L \xrightarrow{\text{nr-rate } O(e^{-|\text{Im } z| T(N_t)/2})} \mathcal{T}_{S^3 \times \mathbb{R}_t}^L$$

identifying the non-compact Lorentzian Dirac as the spectral limit of the framework’s compact-temporal sequence. The propinquity-level closure on the non-compact carrier (single uniform Latrémolière bound to the non-compact target) remains open and is named G2-metric (Paper 47 § 8 Q1).

## 6 Numerical verification and Riemannian-limit recovery

### 6.1 Verification panel

The driver `debug/13b_2_sub_sprint_D_compute.py` builds the joint tunneling pair on the panel

$$(n_{\max}, N_t) \in \{(2, 1), (3, 1), (4, 1), (2, 3), (3, 5), (4, 7)\}$$

at  $T = 2\pi$ . The panel verifies four assertions: (i) Riemannian-limit recovery at  $N_t = 1$ ; (ii) monotone decrease of  $\Lambda^L$  in  $n_{\max}$  at fixed  $N_t$ ; (iii) asymptotic rate consistency  $\Lambda^L(4, 7)/\Lambda^L(2, 3) \approx 0.637$ ; (iv) constituent identities  $\text{reach}_B, \text{reach}_P \leq \gamma^{\text{joint}}$  at every cell.

The numerical results are tabulated in Table 1 (reproduced from `debug/data/13b_2_sub_sprint_D.json`; see Appendix B for full data).

| $(n_{\max}, N_t)$ | $\gamma^{\text{SU}(2)}$ | $\gamma^{U(1)}$        | $C_3^{\text{joint}}$ | cb-norm | $\Lambda^L$ bound | $N_t = 1$ residual   |
|-------------------|-------------------------|------------------------|----------------------|---------|-------------------|----------------------|
| (2, 1)            | 2.0746                  | $\pi/2 \approx 1.5708$ | 0.5774               | 2/3     | 2.0746            | <b>0.0</b> bit-exact |
| (3, 1)            | 1.6101                  | $\pi/2$                | 0.7071               | 1/2     | 1.6101            | <b>0.0</b> bit-exact |
| (4, 1)            | 1.3223                  | $\pi/2$                | 0.7746               | 2/5     | 1.3223            | <b>0.0</b> bit-exact |
| (2, 3)            | 2.0746                  | 0.7220                 | 0.5774               | 2/3     | 2.0746            | —                    |
| (3, 5)            | 1.6101                  | 0.4956                 | 0.7071               | 1/2     | 1.6101            | —                    |
| (4, 7)            | 1.3223                  | 0.3841                 | 0.7746               | 2/5     | 1.3223            | —                    |

Table 1: Numerical verification panel for Theorem 5.3.  $\gamma^{\text{SU}(2)}$  values from Paper 38 § L2 closed-form Stein–Weiss computation;  $\gamma^{U(1)}$  values from standard Fejér-on- $S_T^1$  at  $T = 2\pi$ ;  $C_3^{\text{joint}}$  from the Avery harmonic saturation  $(N - 1)/\sqrt{N^2 - 1}$  on the truncation. The bound on  $\Lambda^L$  is dominated by the SU(2) factor at every cell. The  $N_t = 1$  Riemannian-limit residuals are bit-exact 0.0 in float64, the load-bearing falsifier for the construction.

## 6.2 Riemannian-limit recovery (load-bearing falsifier)

**Proposition 6.1** (Riemannian-limit recovery). *At  $N_t = 1$ , the  $\mathcal{K}^+$ -restricted weak-form Lorentzian propinquity reduces bit-exactly to Paper 38’s SU(2) Riemannian propinquity:*

$$\Lambda^L(\mathcal{T}_{n_{\max}, 1, T}^L, \mathcal{T}_{S^3 \times S_T^1}^L) = \Lambda_{n_{\max}}^{\text{Paper 38}} \quad (\text{Frobenius residual 0.0 in float64}). \quad (34)$$

*Proof.* At  $N_t = 1$ , the only kept Fourier mode is  $q = 0$ , the  $U(1)$  Fejér kernel collapses to the uniform Haar density  $1/T$ , and the temporal Berezin factor  $B^{U(1)}$  is the trivial  $1 \times 1$  identity. By the tensor-product factorization (27),  $B_{n_{\max}, 1, T}^{\text{joint}}(f_s \otimes \mathbf{1}) = B_{\chi\text{-d}}^{\text{SU}(2)}(f_s) \otimes I_1$ , which equals Paper 38’s chirality-doubled spinor lift of the spatial Berezin map.

The chirality-doubled spinor lift on  $\mathcal{K}^+$  is bit-identical to Paper 38’s scalar Fock-basis construction up to the basis representation: the Plancherel weights  $\hat{K}^{\text{SU}(2)}(N)$ , the Lipschitz bound constant  $C_3^{\text{Paper 38}} \leq 1$ , and the convergence rate  $\gamma_{n_{\max}}^{\text{SU}(2)}$  are the same numerical objects. Therefore (34) holds. The bit-exact residual at  $n_{\max} \in \{2, 3, 4\}$  is verified numerically in Table 1.  $\square$

The Riemannian-limit recovery is the load-bearing falsifier of the construction: were it to fail, we would have introduced an extraneous factor in the Wick-rotated signature, miscounted Berezin weights or grids, or had a sign error in the  $\mathcal{K}^+$  restriction. The bit-exact recovery rules out all three. This is the same load-bearing-falsifier pattern used in Paper 42 (Riemannian limit of the Tomita–Takesaki construction at  $N_t = 1$  to Paper 38 / Paper 42 § 11) and Paper 43 (Riemannian limit of the Lorentzian Dirac at  $N_t = 1$  to truthful Camporesi–Higuchi spatial Dirac).

## 6.3 Asymptotic rate consistency

The ratio  $\Lambda^L(4, 7)/\Lambda^L(2, 3) = 1.3223/2.0746 = 0.6374$  is monotone decreasing across the panel, matching the single-factor Paper 38 and Paper 39 [37] ratios bit-identically. This confirms that the SU(2) factor dominates the joint rate at the tested cutoffs, which is the expected behavior given  $\gamma^{\text{SU}(2)} = O(\log n_{\max}/n_{\max})$  and  $\gamma^{U(1)} = O(T/N_t)$ : for  $N_t \gtrsim n_{\max}/\log n_{\max}$ , the SU(2) factor controls the rate.

## 7 Discussion

### 7.1 Four-witness Wick-rotation connection

Paper 42 [39] closes the four-witness Wick-rotation theorem (Hartle–Hawking [12], Sewell [27], Bisognano–Wichmann [2], Unruh [31]) at the operator-system level on the *Riemannian* truncated Camporesi–Higuchi spectral triple, via the geometric BW- $\alpha$  and Tomita–Takesaki BW- $\gamma$  modular Hamiltonians on a hemispheric wedge. Paper 43 [40] extends that closure to signature  $(3, 1)$  on the Krein wedge.

The present paper supplies the Lorentzian *convergence* leg of the arc: Theorem 5.3 shows that the family of truncated Krein spectral triples on which Paper 42/43 work converges in the  $\mathcal{K}^+$ -restricted weak-form propinquity to the continuum reference triple as  $(n_{\max}, N_t) \rightarrow \infty$ . The triple of papers (42, 43, 45) together constitute the Lorentzian-extension closure of the math.OA quartet (38, 39, 40 on the Riemannian side).

### 7.2 Relation to Mondino–Sämman

The synthetic Lorentzian Gromov–Hausdorff program of Mondino–Sämman [20] extends their earlier work [19] to Lorentzian pre-length spaces via  $\epsilon$ -nets of causal diamonds. This is adjacent at the conceptual level (both “Lorentzian Gromov–Hausdorff-type convergence”) but operates on a categorically different mathematical object.

**Remark 7.1** (Different objects, different distances). Mondino–Sämman work on pre-length spaces with causal-diamond geometry; they do not have a Dirac operator or any spectral data, and their convergence is in the synthetic-geometry metric on the space of isometry classes of such pre-length spaces. The present paper works on truncated Krein spectral triples; the convergence is in the operator-algebraic Latrémolière propinquity, restricted to the Krein-positive state-space substrate of the operator system. Neither construction implies the other; both are valid notions of Lorentzian-type Gromov–Hausdorff convergence on their respective categories.

### 7.3 Open questions

- (Q1) **Strong-form Lorentzian propinquity.** A Latrémolière-style metric on Krein-signature spectral triples *without* the  $\mathcal{K}^+$  restriction remains an open NCG-math problem. The technical issue: the standard Latrémolière construction uses the Lipschitz seminorm  $\|[D, \cdot]\|_{\text{op}}$ , which is well-defined on Hilbert space but requires an indefinite-inner-product analog on a Krein space with appropriate Krein-self-adjointness on  $D_L$  per vdD / BBB. The published math.OA literature as of May 2026 does not provide this construction. By performing the  $\mathcal{K}^+$  restriction we trade the strong-form construction for one that uses Hilbert-space machinery on  $\mathcal{K}^+$  and obtains a defensible weak-form convergence theorem at the  $\mathcal{K}^+$ -state-space / Wasserstein–Kantorovich level. Extending to strong form is a multi-month original NCG-math project, and the structural sharpening of Paper 42 § 10 question O3 (the spectral-action-vs-modular-Hamiltonian distinction) suggests the strong form will need additional structural input.
- (Q2) **De-compactification limit  $T \rightarrow \infty$ .** The present construction is at compact temporal radius  $T$ . The  $T \rightarrow \infty$  limit recovers the non-compact  $\mathbb{R}_t$  direction; the joint mass-concentration moment  $\gamma^{U(1)} = O(T/N_t)$  then requires  $N_t$  to grow at least as fast as  $T$ , and the asymptotic rate analysis must be re-examined in this regime. This is internally a separate sprint (L3c in our taxonomy) and is not addressed here.
- (Q3) **Cross-manifold extensions.** The cross-manifold extension  $\mathcal{T}_{S^3} \otimes \mathcal{T}_{\text{Hardy}(S^5)}$  is blocked at the NCG-framework level by the Riemannian-vs-Hardy-sector asymmetry (Paper 24 [34] § V). The four-layer Coulomb/HO asymmetry there (spectrum-computing  $L_0$ , calibration  $\pi$ ,

Wilson gauge with natural matter, modular-Hamiltonian structure of the wedge KMS state) precludes the natural extension within the present framework.

- (Q4) **Sharp asymptotic-tight**  $C_3^{\text{joint}} = 1$ . The present bound has  $C_3^{\text{joint}} \leq 1$  asymptotically tight, with finite-cutoff bound  $\sup_{N \leq 2n_{\max}-1} (N-1)/\sqrt{N^2-1} = \sqrt{1-1/n_{\max}} < 1$  (per Remark 4.5). A direct proof of  $C_3^{\text{joint}} = 1$  in the joint limit (rather than inheriting from the  $\text{SU}(2)$  factor) would require a finer joint analysis of the temporal direction's contribution to the Lipschitz seminorm when restricted to non-pure-tensor multipliers; this is not needed for the qualitative-rate convergence theorem here.
- (Q5) **Quantitative rate constants on the  $U(1)$  factor.** The  $U(1)$  leading constant in  $\gamma^{U(1)} = O(T \log N_t/N_t)$  is standard Cesàro/Fejér; a Stein–Weiss-style derivation of the joint constant matching the  $\text{SU}(2)$  factor's  $4/\pi$  would be a useful refinement, but is not needed for the qualitative-rate convergence theorem.

## 7.4 Structural-skeleton scope

The construction obeys the structural-skeleton scope discipline of the GeoVac framework: the convergence theorem determines the rate, the asymptotic constant, and the limit identification on the  $\mathcal{K}^+$ -substrate at finite cutoff. It does not determine calibration data (Yukawa selection, inner-factor parameters), which are external input. The selection of  $T = 2\pi$  is structurally tied to the Bisognano–Wichmann modular period via Paper 43; the construction itself works for any  $T > 0$ .

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## A Symbolic Plancherel calculation for the joint Fejér symbol

We record the symbolic factorization of the joint Plancherel symbol  $K^{\text{joint}}(j, k)$  used in Lemma 4.1.

**Lemma A.1.** *For all  $n_{\max} \geq 1$ ,  $N_t \geq 1$ , the joint Plancherel symbol factorizes exactly in  $\mathbb{Q}$ :*

$$K^{\text{joint}}_{n_{\max}, N_t}(j, k) = \hat{K}^{\text{SU}(2)}_{n_{\max}}(j) \cdot \hat{K}^{U(1)}_{N_t}(k),$$

with

$$\hat{K}^{\text{SU}(2)}_{n_{\max}}(j) = \frac{2j+1}{Z^{\text{SU}(2)}_{n_{\max}}} \mathbf{1}_{\{j \leq j_{\max}\}}, \quad Z^{\text{SU}(2)}_{n_{\max}} = \frac{n_{\max}(n_{\max}+1)}{2},$$

$$\hat{K}^{U(1)}_{N_t}(k) = \max\left(0, \frac{N_t+1-2|k|}{N_t+1}\right).$$

*Proof.* For the  $\text{SU}(2)$  factor, the spectral Fejér kernel  $K^{\text{SU}(2)} = Z^{-1} |\sum_{j \leq j_{\max}} \sqrt{2j+1} \chi_j|^2$  has Plancherel symbol computed by orthogonality of characters on  $\text{SU}(2)$ :  $\int_{\text{SU}(2)} K^{\text{SU}(2)} \chi_{j'} dg = (2j'+1)/Z$  for  $j' \leq j_{\max}$ , zero otherwise (Paper 38 § L2(a)).

For the  $U(1)$  factor, the Fejér kernel  $K^{U(1)} = N_t^{-1} |D_{N_t}|^2$  where  $D_{N_t} = \sum_{|k| \leq K_{\max}} e^{i 2\pi k \theta / T}$  is the Dirichlet kernel. Direct geometric-series computation gives  $K^{\hat{U}(1)}(k) = \max(0, (N_t+1-2|k|)/(N_t+1))$  (Katznelson [15] Ch. I.2).



The joint kernel is the tensor product on the direct product group; the joint Plancherel symbol on the dual  $\hat{\mathrm{SU}}(2) \times \hat{U}(1)$  factorizes correspondingly via Peter–Weyl on  $\mathrm{SU}(2) \times U(1)$  (direct product gives external tensor products of irreps; characters multiply).  $\square$

## B Numerical verification panel data

The full numerical panel data tabulated in Table 1 is reproduced from `debug/data/l3b_2_sub_sprint_D.json`. The corresponding computational driver is `debug/l3b_2_sub_sprint_D_compute.py`, and the production module is `geovac/lorentzian_propinquity_compact_temporal.py` (approximately 700 lines, with the `LorentzianTunnelingPair` dataclass, the `LorentzianPropinquityBound` dataclass, the `compute_lorentzian_propinquity_bound` function, and the `verify_riemannian_limit_at_N_t` bit-exact check).

The numerical verification of all constituents in Proposition 5.2 is in `tests/test_lorentzian_propinquity.py`. (15+ test functions covering the tunneling pair construction, reach and height bounds, the main theorem panel, the asymptotic rate consistency, the  $\mathcal{K}^+$  restriction correctness, the Riemannian-limit recovery, and the monotone decrease in  $n_{\max}$ ).

*Convergence diagnostic.* The ratio  $\Lambda^L(4, 7)/\Lambda^L(2, 3) = 1.3223/2.0746 = 0.6374$  matches the single-factor Paper 38 / Paper 39 ratio bit-identically, confirming that the  $\mathrm{SU}(2)$  factor dominates the joint rate at the tested cutoffs. At larger cutoffs the  $U(1)$  factor would re-enter at order  $O(T/N_t)$  if  $N_t$  does not grow as fast as  $n_{\max}/\log n_{\max}$ .

*Memory note.* The panel cell  $(n_{\max}, N_t) = (5, 9)$  was originally planned but exceeds the operator-system vec-matrix memory budget at  $\dim_K = 1260$  and  $K = 2565$  multipliers ( $\approx 60$  GiB `complex128`). The panel is restricted to  $\{(2, 3), (3, 5), (4, 7)\}$ , which is sufficient for Riemannian-limit recovery, monotone-decrease, and asymptotic-rate consistency. Larger cells would require sparse linear-algebra refactoring of `geovac.operator_system_compact_temporal`, which is a downstream engineering item and not an analytical blocker.

## References

- [1] B. Bekka, P. de la Harpe, and A. Valette, *Kazhdan’s Property (T)*, New Math. Monographs **11**, Cambridge University Press, 2008.
- [2] J. J. Bisognano and E. H. Wichmann, *On the duality condition for quantum fields*, J. Math. Phys. **17** (1976), 303–321.
- [3] N. Bizi, C. Brouder, and F. Besnard, *Space and time dimensions of algebras with application to Lorentzian noncommutative geometry and quantum electrodynamics*, J. Math. Phys. **59** (2018), 062303; arXiv:1611.07062.
- [4] M. Bożejko and G. Fendler, *Herz–Schur multipliers and uniformly bounded representations of discrete groups*, Arch. Math. **57** (1991), 290–298.
- [5] R. Camporesi and A. Higuchi, *On the eigenfunctions of the Dirac operator on spheres and real hyperbolic spaces*, J. Geom. Phys. **20** (1996), 1–18; arXiv:gr-qc/9505009.
- [6] A. Connes and W. D. van Suijlekom, *Spectral truncations in noncommutative geometry and operator systems*, Comm. Math. Phys. **383** (2021), 2021–2067; arXiv:2004.14115.
- [7] A. Devastato, F. Lizzi, and P. Martinetti, *Time as an emergent property in spectral physics: gauge theories from spectral noncommutative geometry*, J. Math. Phys. **59** (2018), 112301.
- [8] C. Farsi and F. Latrémolière, *Continuity of spectral propinquity for spectrally truncated spheres*, preprint, 2024.

- [9] C. Farsi and F. Latrémolière, *Continuity for the spectral propinquity of the Dirac operators associated with an analytic path of Riemannian metrics*, arXiv:2504.11715, 2025.
- [10] N. Franco and M. Eckstein, *An algebraic formulation of causality for noncommutative geometry*, Class. Quantum Grav. **30** (2013), 135007; arXiv:1212.5171.
- [11] R. Geroch, *Topology in general relativity*, J. Math. Phys. **8** (1967), 782–786.
- [12] J. B. Hartle and S. W. Hawking, *Path-integral derivation of black-hole radiance*, Phys. Rev. D **13** (1976), 2188–2203.
- [13] E. Hekkelman and E. McDonald, *Spectral truncations of  $\mathbb{T}^d$  and quantum metric geometry*, J. Noncommut. Geom., to appear (2024); arXiv:2403.18619.
- [14] E. Hekkelman and E. McDonald, *A noncommutative integral on spectrally truncated spectral triples, and a link with quantum ergodicity*, J. Funct. Anal., to appear (2025); arXiv:2412.00628.
- [15] Y. Katznelson, *An Introduction to Harmonic Analysis*, 3rd ed., Cambridge University Press, 2004.
- [16] F. Latrémolière, *The Gromov–Hausdorff propinquity for metric spectral triples*, Adv. Math. **415** (2023), Paper No. 108876, 88pp; preprint arXiv:1811.10843, November 2018.
- [17] F. Latrémolière, *The Gromov–Hausdorff propinquity*, Trans. Amer. Math. Soc. **370** (2018), 365–411.
- [18] L. Leimbach and W. D. van Suijlekom, *Gromov–Hausdorff convergence of spectral truncations of the torus*, Adv. Math. **439** (2024), Paper No. 109496.
- [19] A. Mondino and C. Sämann, *Lorentzian metric measure spaces*, Lett. Math. Phys. **114** (2024), Paper No. 37; arXiv:2209.14384.
- [20] A. Mondino and C. Sämann, *Synthetic Lorentzian Gromov–Hausdorff convergence and pre-compactness*, arXiv:2504.10380, 2025.
- [21] M. Che, R. Perales, and C. Sormani, *Gromov’s compactness theorem for the intrinsic timed-Hausdorff distance*, arXiv:2510.13069, October 2025.
- [22] G. Nieuviarts, *Emergence of Lorentz symmetry from an almost-commutative twisted spectral triple*, arXiv:2502.18105 (v3, May 2025).
- [23] G. Nieuviarts, *Emergence of time from a twisted spectral triple in almost-commutative geometry*, arXiv:2512.15450, December 2025 (revised May 2026).
- [24] V. I. Paulsen, *Completely Bounded Maps and Operator Algebras*, Cambridge Studies in Advanced Mathematics 78, Cambridge University Press, 2002.
- [25] J.-P. Pier, *Amenable Locally Compact Groups*, Wiley, 1984.
- [26] G. Pisier, *Similarity Problems and Completely Bounded Maps*, Lecture Notes in Math. 1618, Springer, 2nd ed., 2001.
- [27] G. L. Sewell, *Quantum fields on manifolds: PCT and gravitationally induced thermal states*, Ann. Phys. (NY) **141** (1982), 201–224.
- [28] A. Strohmaier, *On noncommutative and pseudo-Riemannian geometry*, J. Geom. Phys. **56** (2006), 175–195.

- [29] M. Takesaki, *Theory of Operator Algebras I*, Springer, 1979.
- [30] M. Toyota, *Quantum Gromov–Hausdorff convergence of spectral truncations for groups with polynomial growth*, arXiv:2309.13469, 2023.
- [31] W. G. Unruh, *Notes on black-hole evaporation*, Phys. Rev. D **14** (1976), 870–892.
- [32] K. van den Dungen, *Krein spectral triples and the fermionic action*, Math. Phys. Anal. Geom. **19** (2016), Art. 4, 22pp; arXiv:1505.01939.
- [33] J. de Groot, *Pseudo-Riemannian Spectral Triples for  $SU(1,1)$* , arXiv:2601.22171, January 2026.
- [34] J. Loutey, *Bargmann–Segal lattice for the 3D harmonic oscillator and the Coulomb/HO structural asymmetry*, GeoVac internal preprint (2026), DOI via Zenodo.
- [35] J. Loutey, *The GeoVac spectral triple, the master Mellin engine case-exhaustion theorem, and the GH-convergence theorem on  $S^3$* , GeoVac internal preprint, papers/synthesis (2026), DOI via Zenodo.
- [36] J. Loutey, *Latrémolière propinquity convergence of truncated Camporesi–Higuchi spectral triples on  $S^3$* , GeoVac internal preprint, papers/standalone (2026), DOI via Zenodo.
- [37] J. Loutey, *Tensor-product propinquity convergence for distinct-focal-length truncated Camporesi–Higuchi spectral triples on  $S^3$* , GeoVac internal preprint, papers/standalone (2026), DOI via Zenodo.
- [38] J. Loutey, *Latrémolière propinquity convergence of spectral truncations of compact Lie groups with bi-invariant metric*, GeoVac internal preprint, papers/standalone (2026), DOI via Zenodo.
- [39] J. Loutey, *Tomita–Takesaki modular structure on truncated  $SU(2)$  spectral triples; four-witness Wick-rotation literal identification at finite cutoff*, GeoVac internal preprint, papers/standalone (2026), DOI via Zenodo.
- [40] J. Loutey, *Lorentzian extension of the four-witness Wick-rotation theorem at finite cutoff*, GeoVac internal preprint, papers/standalone (2026), DOI via Zenodo.
- [41] J. Loutey, *Operator-system extension of the BBB Krein spectral triple at finite cutoff: propagation number and envelope dependence on the Camporesi–Higuchi spinor bundle*, GeoVac internal preprint, papers/standalone (2026), DOI via Zenodo.
- [42] J. Loutey, *Strong-form Lorentzian propinquity convergence on truncated  $SU(2) \times U(1)_T$  Krein spectral triples*, GeoVac internal preprint, papers/standalone (2026), DOI via Zenodo. (Companion paper closing the strong-form gap identified as Q1/G1 of the present paper; identified the envelope refinement to (21) in Remark 4.5.)
- [43] J. Loutey, *Norm-resolvent convergence to the non-compact Lorentzian Krein spectral triple on  $S^3 \times \mathbb{R}_t$  and the two-rate hybrid limit*, GeoVac internal preprint, papers/standalone (2026), DOI via Zenodo. (Closes the de-compactification limit G2 of the present paper at the norm-resolvent / spectral level via a two-rate hybrid composite; propinquity / metric level on a non-compact carrier remains open as named G2-metric / G1-genuine.)